Answer all questions

For Examiner's Use

1 **(a)** Name the type of intermolecular forces found in

(i) chloromethane,

____

(ii) bromine,

____

(iii) water.

______________________________ [3]

(b) **Table 1.1** shows the boiling points of propanone and butane.

	boiling point/K
propanone	329
butane	273

Explain the difference in boiling points.

____

____

____

______________________________ [2]

(c) **(i)** Describe the bonding in graphite.

____

____

(ii) Explain any two physical properties of graphite.

1. ______________________________

____

2. ______________________________

____

(iii) State any one use of graphite.

______________________________________ [5]

[Total: 10]

2 (a) Define the term

(i) activation energy.

____

____

(ii) homogenous catalyst.

____

______________________________________ [2]

(b) Methanol can be synthesised industrially from a mixture of carbon monoxide and hydrogen using zinc oxide and chromium (III) oxide as catalysts. The enthalpy of the reaction is $-90.8\ KJmol^{-1}$.

(i) Write a balanced equation for the synthesis of methanol.

(c) Table 2.1 shows combustion data for hydrogen, carbon and carbon monoxide

Table 2.1

	ΔH_c^{θ}/KJmol^{-1}
carbon	–394
hydrogen	–286
carbon monoxide	–284

Calculate the enthalpy change of combustion of methanol.

[2]

(d) On combustion, 2 g of methanol increased the temperature 100 g of water from 31°C to 65°C.

Calculate the enthalpy change of combustion of methanol.

[3]

[Total: 10]

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For Examiner's Use

3 (a) Fig. 3.1 shows some reactions of magnesium and its compounds.

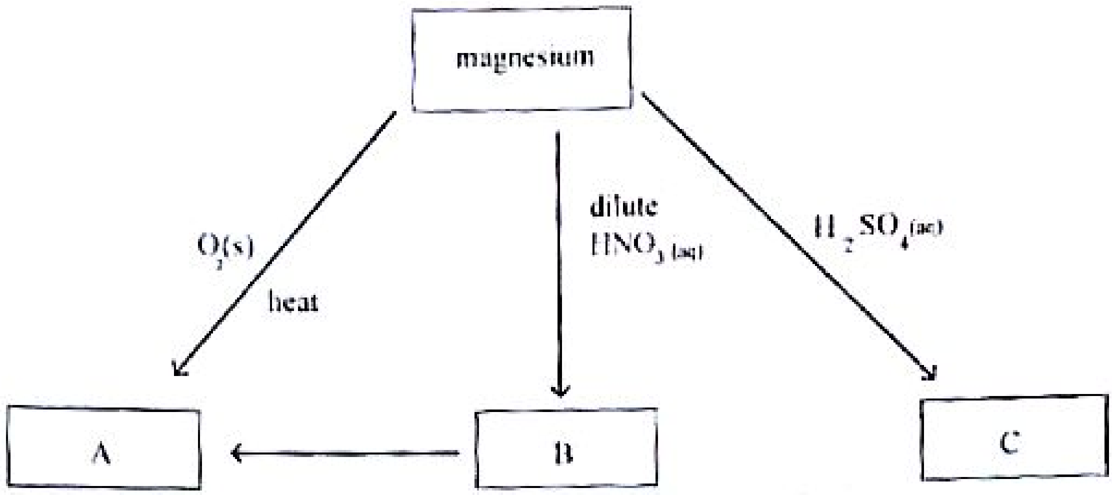


Fig. 3.1

(i) Describe and explain how compound **B** can be treated to form compound **A**.

____

____

(ii) Write a balanced chemical equation for conversion of **B** to **A**.

____

(iii) Suggest why compound **B** is suitable for the development of efficient heat transfer and thermal storage fluid systems in industry.

____

____

____

____

For Examiner's Use

(b) The effects of treating soil samples with compound **A** and **C** were investigated.

(i) Describe the effects of treating a soil sample with

1. compound **A**,

2. compound **C**.

(ii) Describe and explain how the mineral composition of a soil sample treated with compound **A** differs from that treated with compound **C**.

[5]

[Total:10]

4 Fig. 4.1 shows how compound **Y** can be synthesised.

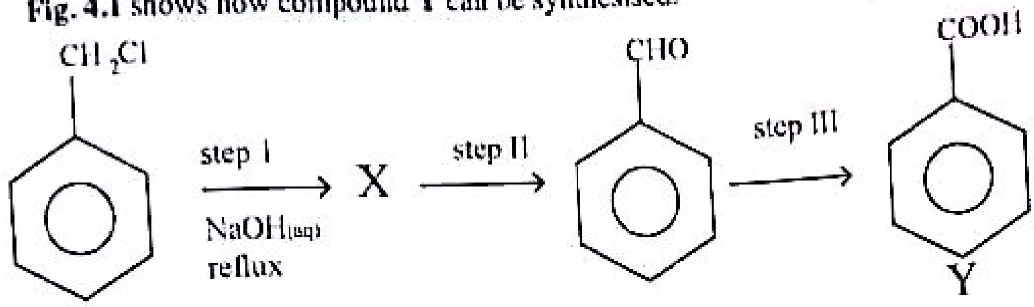


Fig. 4.1

(a) **(i)** Draw the structural formula of **X**.

(ii) State the type of reaction in

1. step **I**.

____

2. step **II**.

____

(iii) State the reagents and conditions for step II

reagents ______________________________

conditions ______________________________

______________________________ [5]

(b) Describe and explain the observations made in step **II**.

____

____

____

______________________________ [2]

(c) Describe a simple test tube reaction to distinguish between **X** and **Y**

[3]

[Total: 10]

5 (a) (i) Define an *electrophile*

(ii) Describe a mechanism for the electrophilic addition of bromine to cyclohexene

For Examiner's Use

(b) Propane reacts with chlorine to form two different monochloropropanes.

(i) Draw the structural formulae of the two monochloropropanes.

(ii) State the type of isomerism shown in **(b)(i)**.

____

(iii) Write the reagents and conditions for the hydrolysis of any one of the monochloropropanes.

____

____

(iv) Draw the displayed formula of the organic product in **(b)(iii)**.

[6]
[Total: 10]

[Turn over

6 (a) (i) Name the major ores from which platinum is extracted.

1. ____________________

2. ____________________ [2]

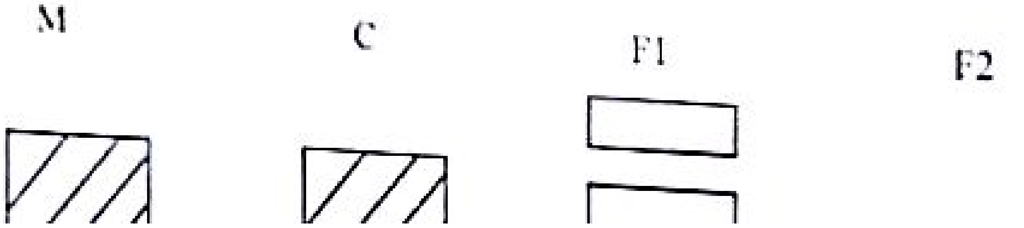
(ii) State any **three** properties of platinum which makes the mineral ideal for jewellery.

____

____

____________________ [3]

(b) Fig. 6.1 shows results from electrophoresis of DNA of a child(C), its mother (M) and two possible fathers, F1 and F2.



For Examiner's Use

(i) State with a reason whether the child is related to

1. father F2.

____

____

2. the mother, M

____

____ [4]

(ii) State any other field apart from medicine that DNA finger printing can be used.

____ [1]

[Total:10]